

## HOMEWORK 2

### *Track 2*

**Due date:** October 16, 2023.

**Reading assignment:** Finish reading chapter 2 of the class notes (on class web-site). Finish reading chapters 1, 2 & 3 of the text book. Read chapter 3 of the class notes. Remember to include the self-assessment grades on the first sheet of your home work (see handout for details).

1. Show that any matrix  $A$  can be factored as  $A = XY^T$ , where  $X$  and  $Y$  have full column-rank.
2. Let  $x$  and  $y$  be real column vectors. Give an explicit  $LU$  factorization of the matrix  $xy^T$ . What is its rank?
3. Suppose

$$A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix},$$

with  $A$  and  $A_{11}$  square non-singular matrices. Then

$$S = A_{22} - A_{21}A_{11}^{-1}A_{12}$$

is a Schur complement of  $A$ . Show that

$$A^{-1} = \begin{pmatrix} * & * \\ * & S^{-1} \end{pmatrix}.$$

In other words  $S^{-1}$  is the bottom-right sub-matrix of  $A^{-1}$ .

4. Suppose  $A \in \mathbb{R}^{n \times n}$  and  $b \in \mathbb{R}^{n \times 1}$ , and  $\varphi(x) = \frac{1}{2}x^T A x - x^T b$ . Show that the gradient of  $\varphi$  is given by  $\nabla \varphi(x) = \frac{1}{2}(A^T + A)x - b$ .
5. Let  $A = \begin{pmatrix} 1/3 & 1/3 & 1/3 \end{pmatrix}^T$ . Think of  $A$  as a map from  $\mathbb{R}^1$  to  $\mathbb{R}^3$  via matrix-vector multiplication. Show that the map is one-to-one. Find two linear left-inverses for  $A$ . Find a left-inverse for  $A$  that is *not* linear.
6. Find all matrices  $X$  that satisfy the equation  $A X B^T = C$ , in terms of the  $LU$  factorization of  $A$  and  $B$ . When are there no solutions?
7. Let  $U_1$  and  $U_2$  be two upper-triangular matrices and let  $Z$  be an  $m \times n$  matrix. Let  $X$  be an unknown matrix that satisfies the equation  $U_1 X + X U_2 = Z$ .
  - A) Give an algorithm that finds  $X$  in  $O(mn(m+n))$  arithmetic operations.
  - B) Find conditions on  $U_1$  and  $U_2$  that guarantee the existence of a unique solution  $X$ .
  - C) Give a non-trivial example ( $U_1 \neq 0$ ,  $U_2 \neq 0$  and  $X \neq 0$ ) where those conditions are not satisfied and  $U_1 X + X U_2 = 0$ .